

Perversions in Helices: Phase Transition and Soliton Cristalization

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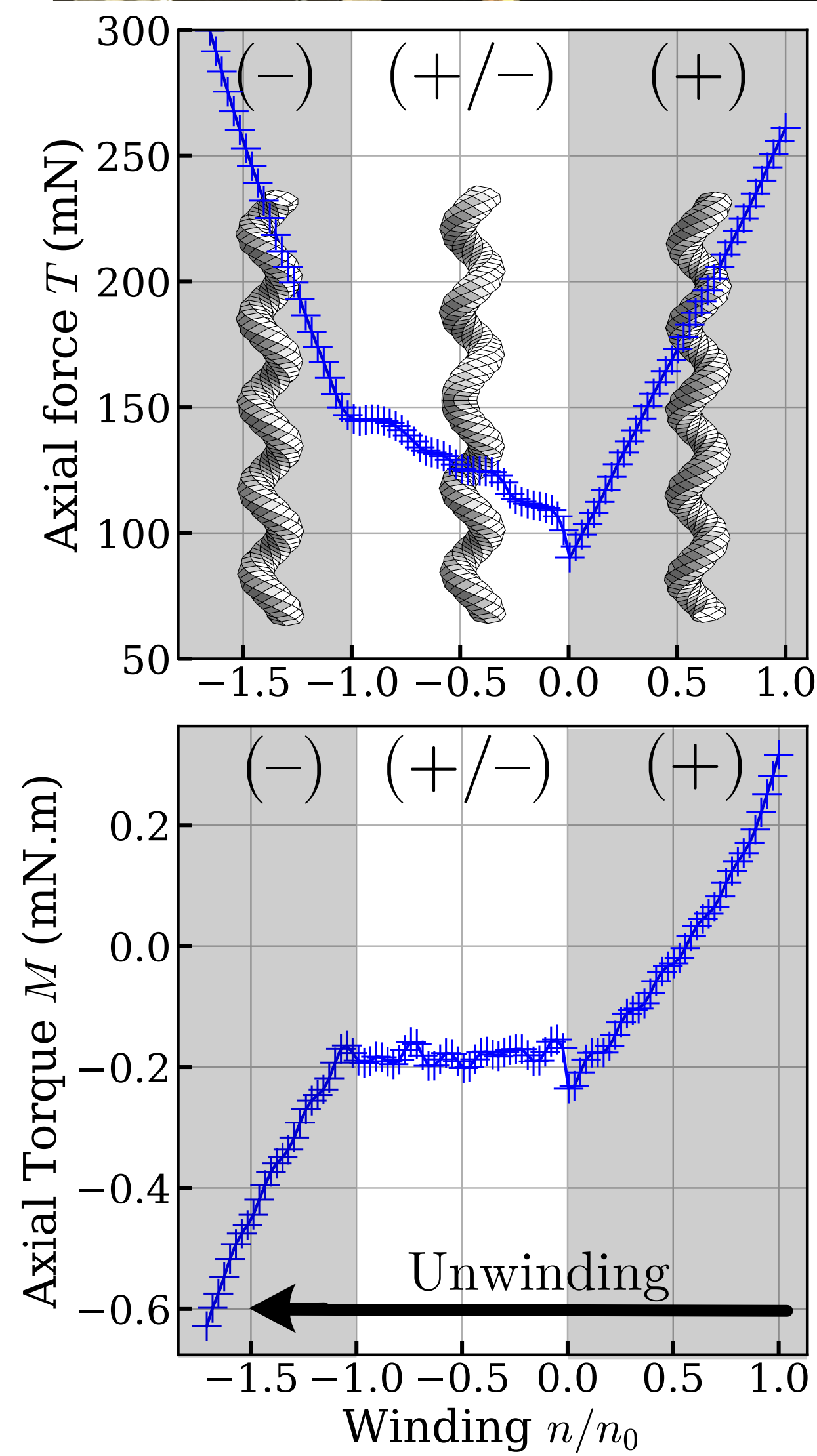
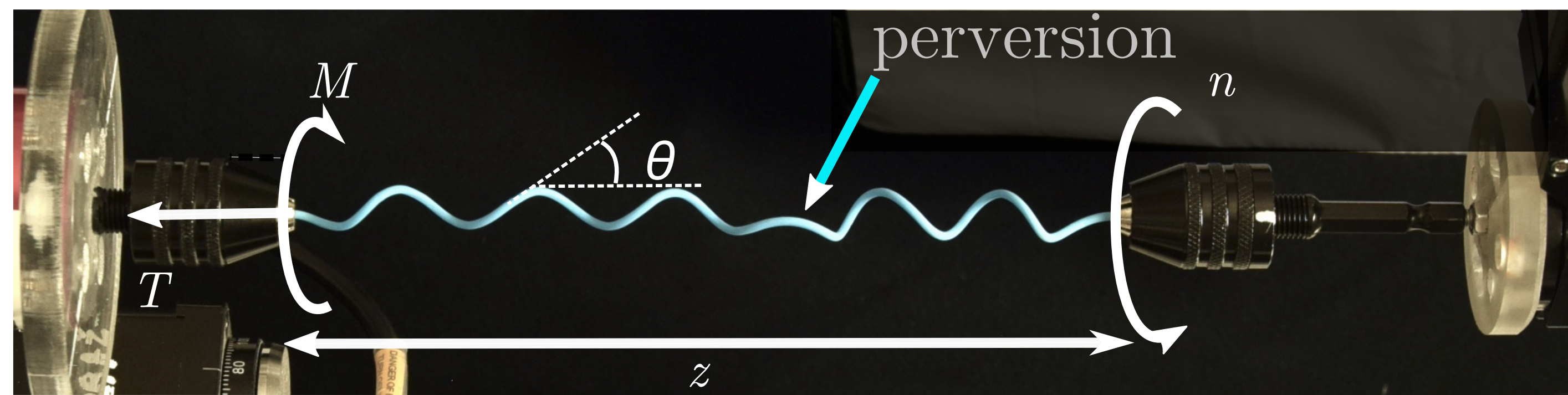
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1. Introduction

The interest regarding elastic rods was renewed in recent fields, such as molecular biology, plant development and softrobotics [1, 2, 3]. Our work focuses on helices, helical unwinding and, in particular, on helix handedness reversal with the formation of a *perversion*.

2. Unwinding of Helix



Typical unwinding experiment: A helical rod is clamped at both ends, one end rotates to unwind it (test it with the rod !).

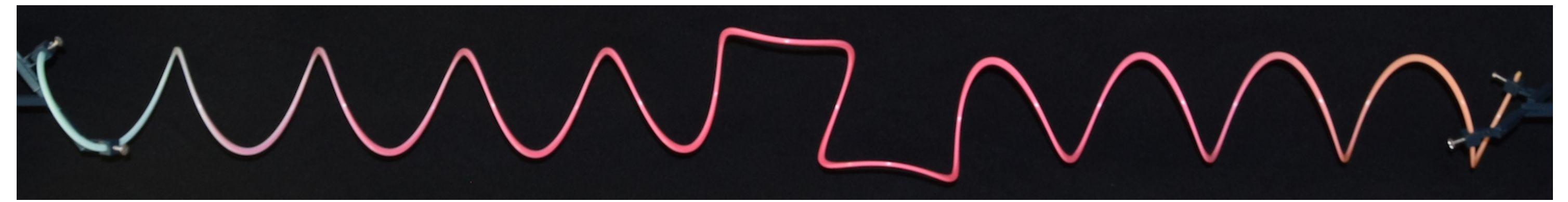
(+): unwinding of the pure helix of natural chirality

(+/-): coexistence of a helix of natural chirality and opposite chirality connected by a perversion. The axial torque is fixed. The perversion propagates along the rod.

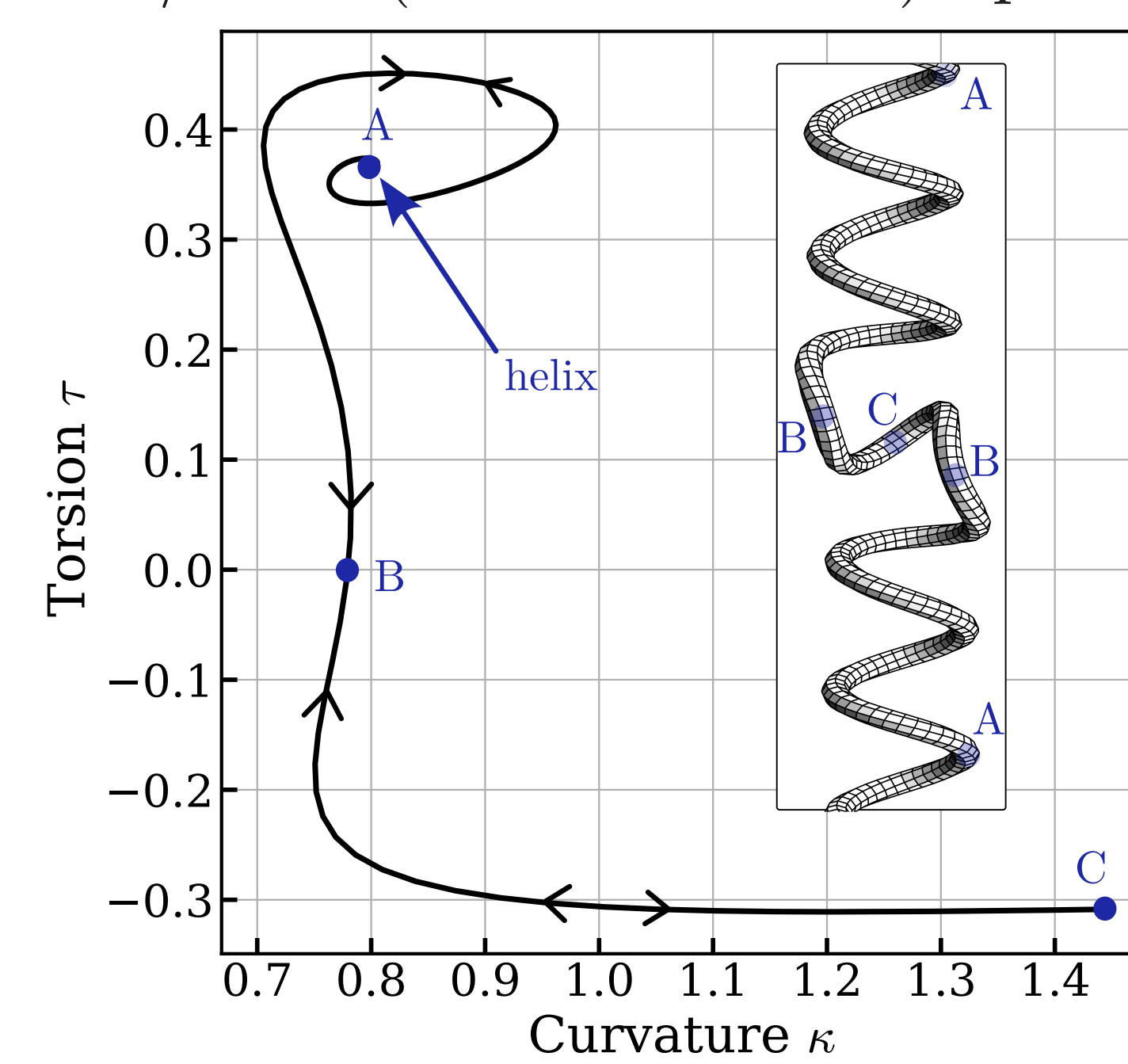
(-): overwinding of the pure helix of opposite chirality.

4. Interaction of perversions

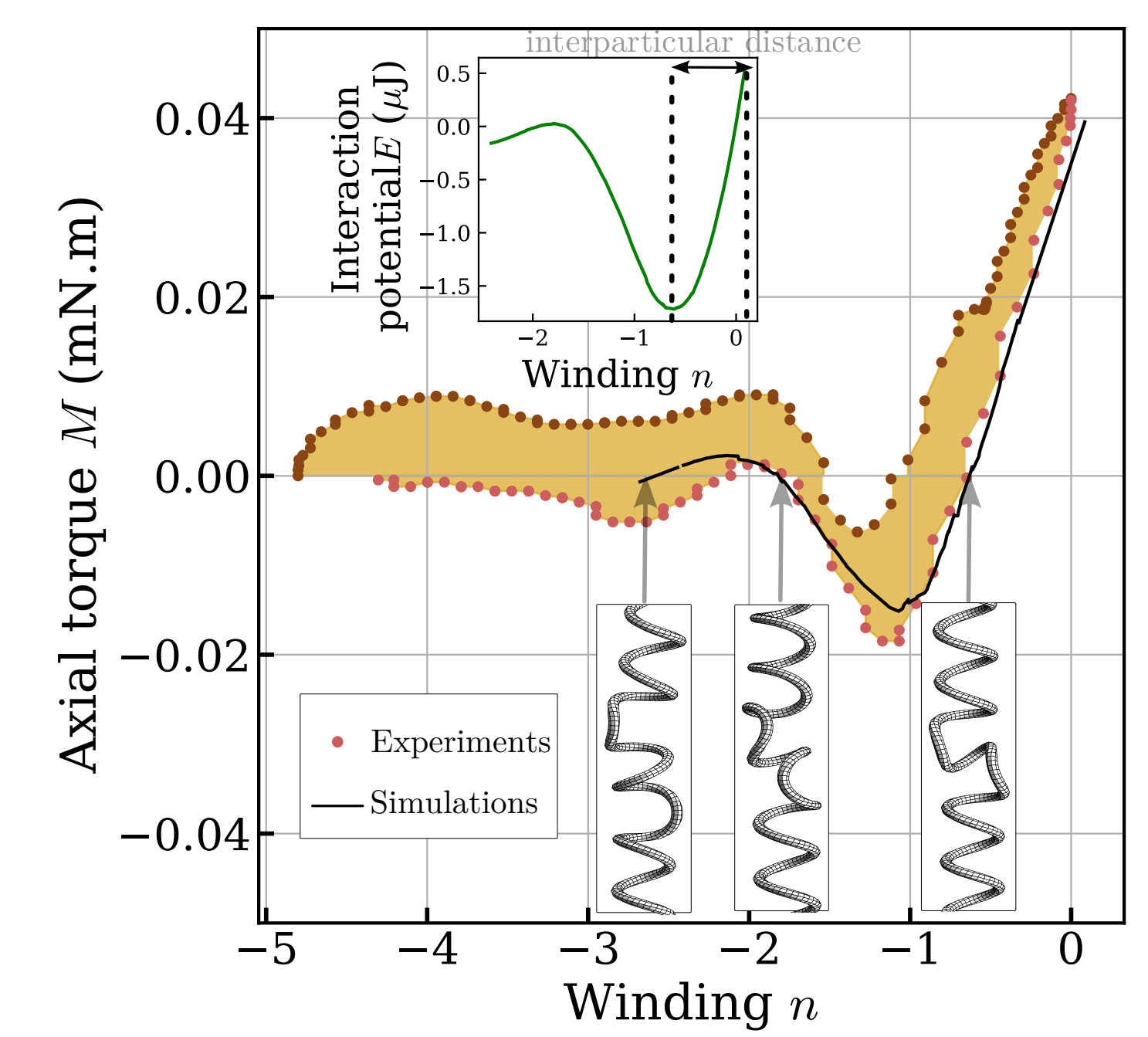
Perversions, as solitons, can propagate along the rod without deformation. Two perversions act as interacting particles. If one end is free to rotate, there is an equilibrium distance between the particles.



Numerical solution in curvature torsion space of a perversion/anti-perversion configuration obtained by shooting method in the curvature/force (6 dimensionnal) space.

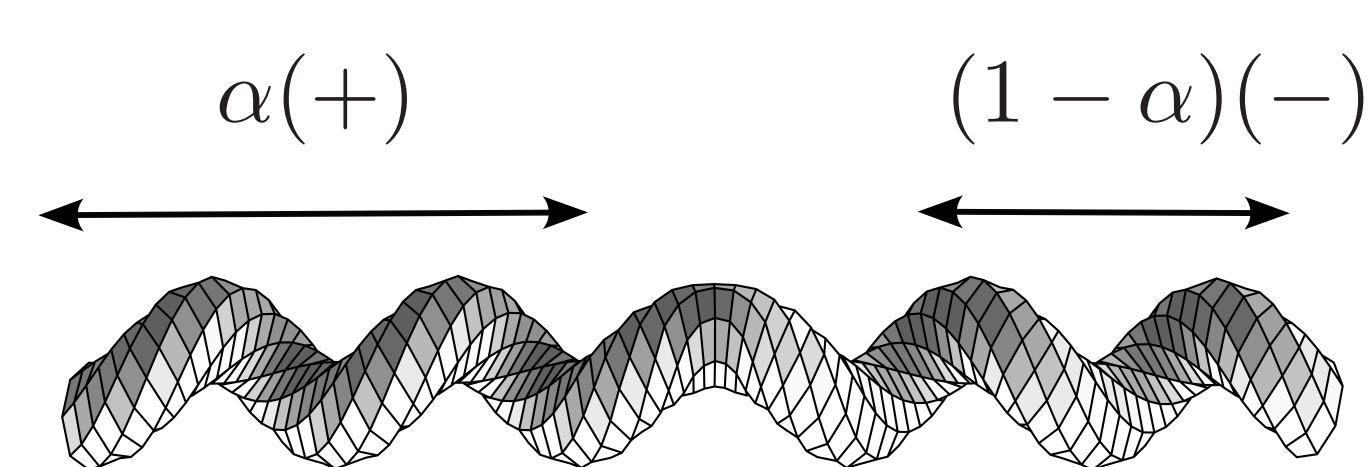


Axial torque as a function of the number of turns applied on one end and corresponding interaction potential of the perversions.



3. Model: Phase Transition

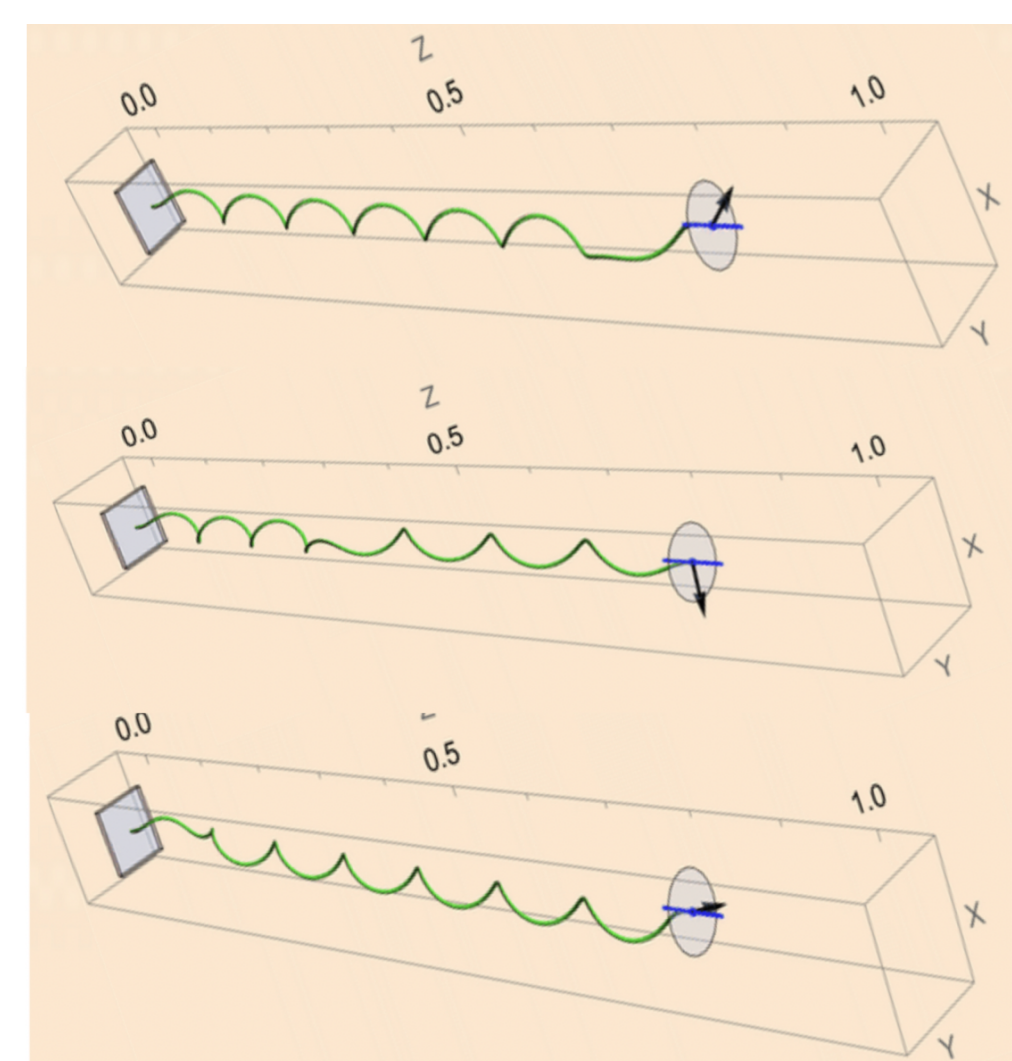
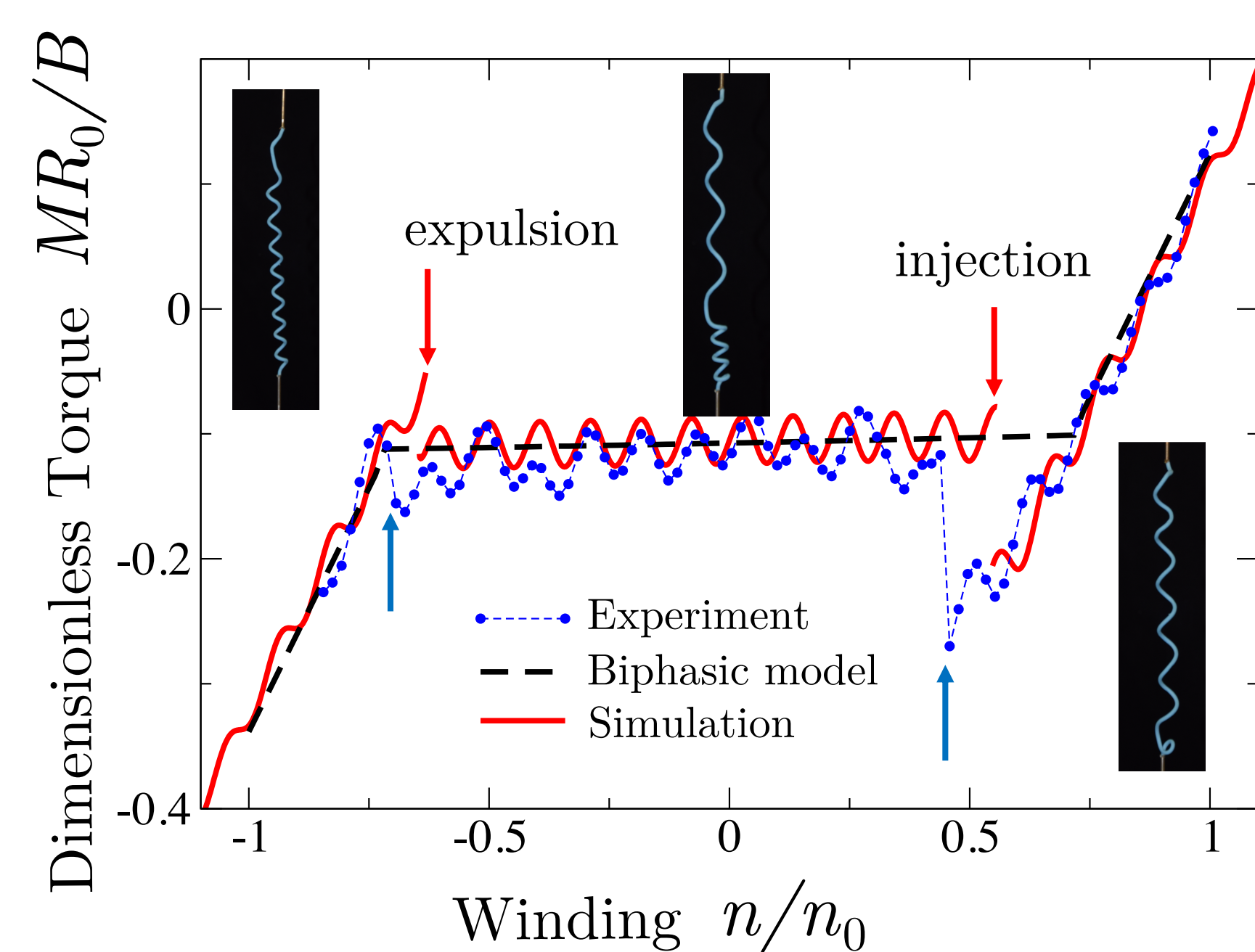
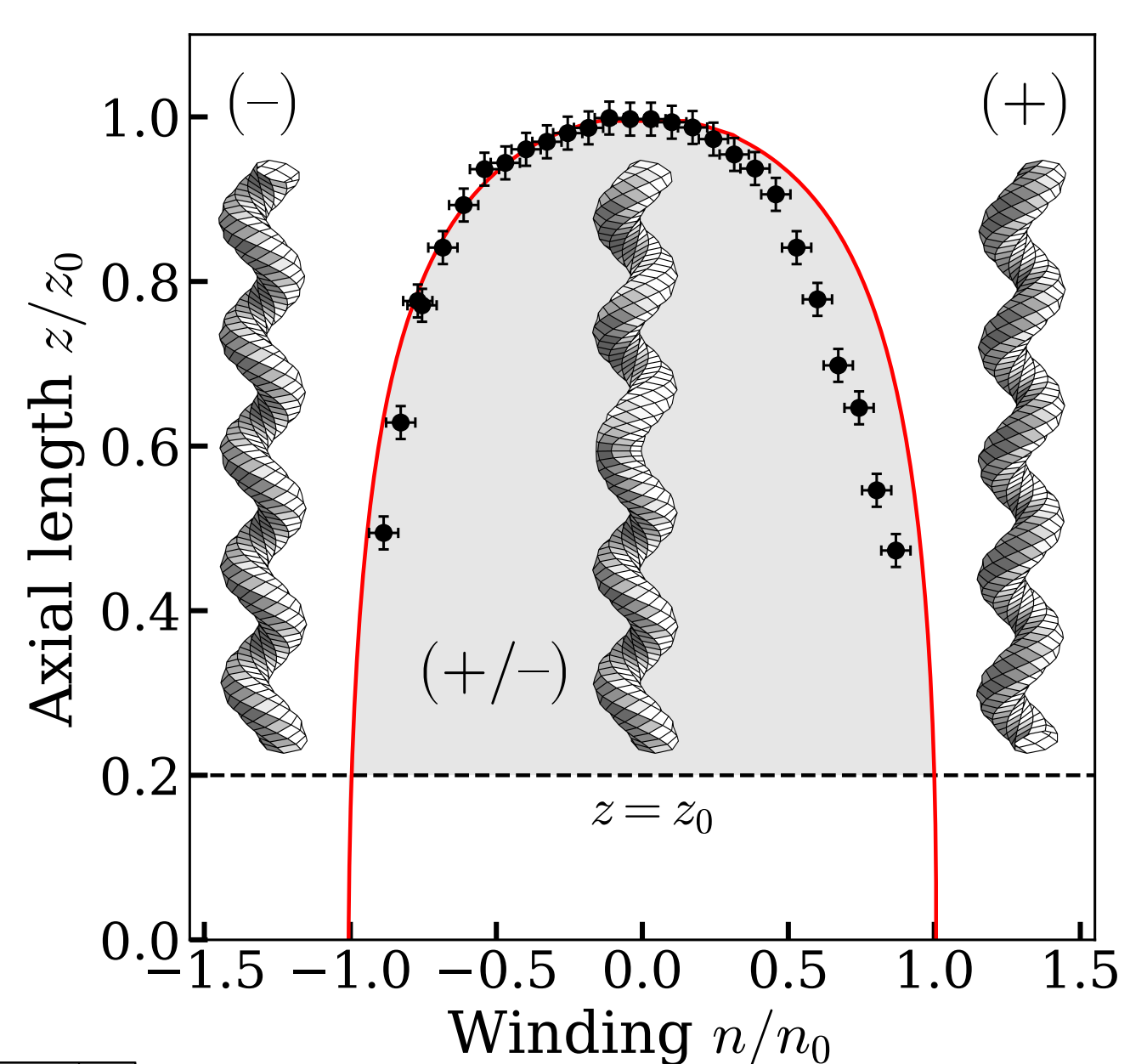
Expansion of the energy for a Kirchhoff rod composed of two phases:



$$E = \alpha E_+ + (1 - \alpha) E_-$$

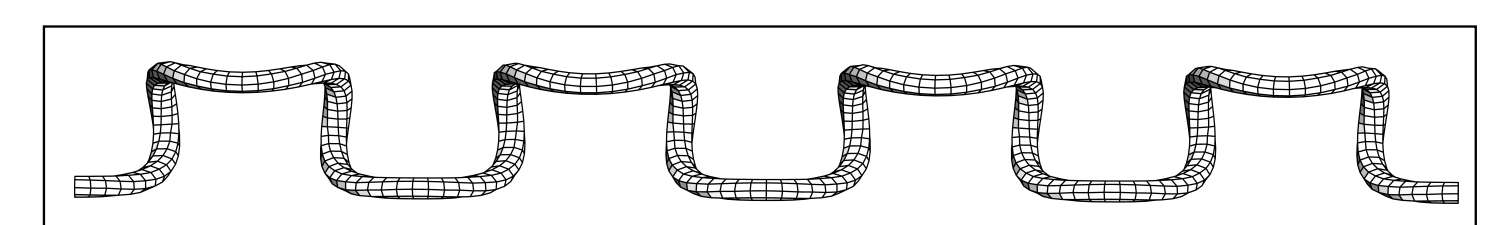
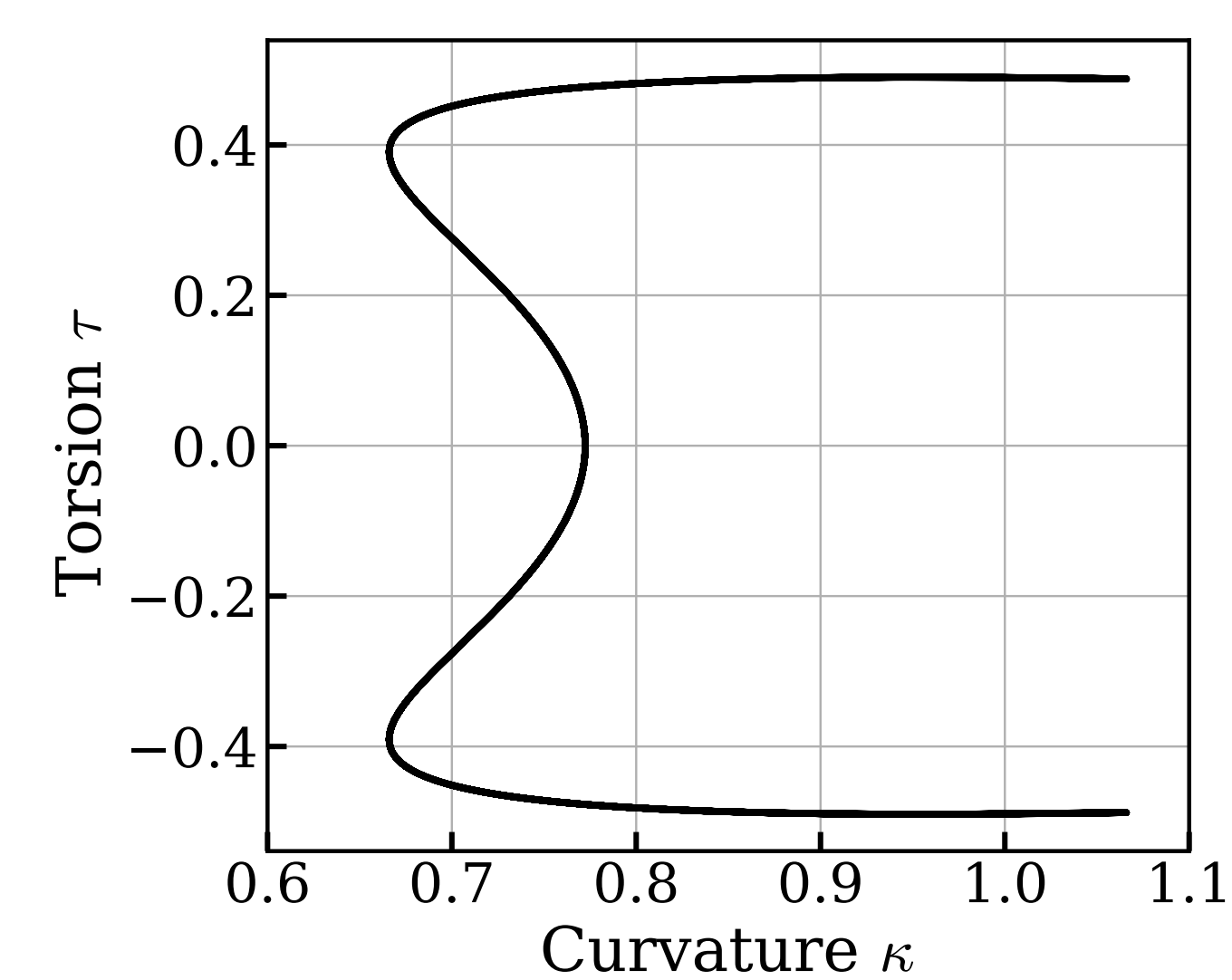
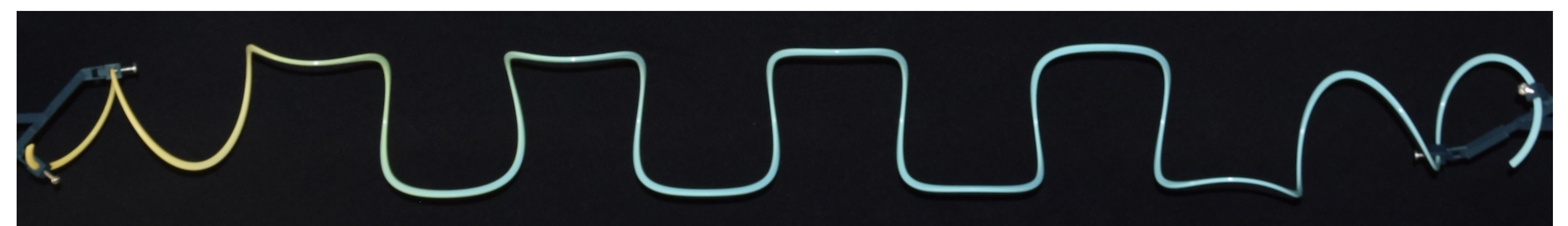
$$E_{\pm} = \frac{1}{2} (\kappa_{\pm} - \kappa_0)^2 + \frac{1}{2} \Gamma (\tau_{\pm} - \tau_0)^2$$

Corresponding phase diagram for different extensions



5. Perversion lattice

There is an interparticle equilibrium distance: we can bond several particles.



Numerical solution in curvature torsion space and real space of a periodic configuration.

6. Conclusions

The unwinding of helix gives rise to the formation of a perversion and an helix of opposite chirality. This can be rationalized in terms of a phase coexistence model. The perversions act as interacting particles presenting an equilibrium interparticle distance; making it possible to create periodic configurations.

References

- [1] Bryant, Z., Stone, M. D., Gore, J., Smith, S. B., Cozzarelli, N. R., and Bustamante, C. (2003) Structural transitions and elasticity from torque measurements on DNA. *Nature*, **424**(6946), 338–341.
- [2] McMillen and Goriely (2002) Tendril Perversion in Intrinsically Curved Rods. *Journal of Nonlinear Science*, **12**(3), 241–281.
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